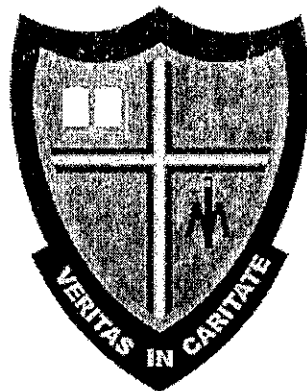




ST BENEDICT'S

SUBJECT	Mathematics	PAPER	Algebra Part A
GRADE	11	DATE	13 February 2019
EXAMINER	Miss A Balzer	MARKS	35
NAME		MODERATOR	Gr 11 Teachers
TEACHER	MEMO	DURATION	40min

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Equations	35	

QUESTION 1

35 MARKS

Solve for x :

a. $x^2 - 9x + 20 = 0$

(3)

$$(x-4)(x-5) = 0$$

$$x = 4 \text{ or } x = 5$$

b. $3x^2 + 5x = 4$ Correct to two decimal places

(4)

$$3x^2 + 5x - 4 = 0 \quad \checkmark \text{ std form}$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-4)}}{2(3)} \quad \checkmark \text{ sub.}$$

$$= \frac{-5 \pm \sqrt{73}}{6}$$

$$x = -2,26 \text{ or } x = 0,59$$

c. $\frac{x+1}{x+3} + \frac{x+5}{x-2} = \frac{14x+7}{x^2+x-6}$

(8)

$$\frac{x+1}{x+3} + \frac{x+5}{x-2} = \frac{14x+7}{(x+3)(x-2)}$$

LCD: $(x+3)(x-2)$

Rest: $x \neq -3$

$x \neq 2$

$$(x+1)(x-2) + (x+5)(x+3) = 14x+7$$

$$x^2 - x - 2 + x^2 + 8x + 15 = 14x + 7$$

$$2x^2 - 7x + 6 = 0$$

$$(2x-3)(x-2) = 0$$

$$x = 3/2 \text{ or } x = 2$$

N/A.

f. $x^2 - 8x = 13$ by completing the square and leaving answers in simplest surd form
(6)

$$x^2 - 8x - 13 = 0$$

$$x^2 - 8x + 16 - 13 - 16 = 0$$

$$(x - 4)^2 - 29 = 0$$

$$(x - 4)^2 = 29$$

$$x - 4 = \pm \sqrt{29}$$

$$x = 4 \pm \sqrt{29}$$